

CST8513: Quality Assurance and Testing

Assignment III (Phase 3: Automation Testing)

Deliverable 3: Final Automation Test Report

Target Web Application: SauceDemo (Swag Labs) - <https://www.saucedemo.com/>

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Course: CST8513 – Quality Assurance and Testing

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GitHub Repository: <https://github.com/MeetAhalpara/QA-Test-Automation>

Table of Contents

CST8513: Quality Assurance and Testing	1
Assignment III (Phase 3: Automation Testing)	1
Deliverable 3: Final Automation Test Report	1
1. Executive Summary.....	3
Project Evolution Across Three Phases:	3
Key Automation Outcomes:	3
2. Test Objectives & Scope	4
2.1 Primary Objectives.....	4
2.2 In-Scope Functional Modules	4
3. Automation Framework Architecture (Page Object Model)	5
4. Test Execution Summary & Detailed Results	7
4.1 Quantitative Results Overview	7
4.2 Comprehensive Traceability Matrix & Execution Details.....	7
5. Continuous Integration (CI/CD) Implementation	12
5.1 Jenkins Pipeline (ci_cd/Jenkinsfile).....	12
5.2 GitHub Actions Workflow (.github/workflows/automation-tests.yml)	12
6. Discrepancies & Comparison with Phase 2 (Manual Testing).....	13
7. Analysis of Phase 2 Defects in Automation	14
7.1 Defect D-01: Empty Cart Checkout (Severity: High)	14
7.2 Defect D-02: Weak Input Validation (Severity: Medium)	14
8. Conclusion & References	15
8.1 Project Conclusion	15
8.2 Technical References & Artifacts	15
References	15

1. Executive Summary

This Final Automation Test Report represents the formal culmination of Assignment III (Phase 3: Automation Testing) for the SauceDemo (Swag Labs) e-commerce web application in CST8513.

Project Evolution Across Three Phases:

1. Phase 1 (Application Proposal & Requirements Analysis):
Analyzed the application architecture, business flows, and defined 14 core functional requirements (FR-01 through FR-14) encompassing authentication, product catalog browsing, dynamic sorting, product specifications, cart management, checkout forms, financial tax calculations, order completion, and secure logout.
2. Phase 2 (Manual Test Execution & Defect Reporting):
Authored and executed 14 manual functional test cases (TC-01 through TC-14), achieving an 85.7% pass rate (12 passed, 2 failed). Two functional defects were formally documented:
 - Defect D-01 (High Severity): Empty Cart Checkout Allowed — Users can proceed to checkout with zero products in the cart.
 - Defect D-02 (Medium Severity): Weak Customer Information Input Validation — Checkout forms accept single-character inputs without format constraints.
3. Phase 3 (Automation Framework & Test Implementation - Current Phase):
Successfully developed an automated regression testing suite using Python 3.11, Selenium WebDriver structured under the Page Object Model (POM) design pattern, Pytest, and Robot Framework.

Key Automation Outcomes:

- Total Test Scenarios Automated: 22 Selenium test scenarios + 14 Robot Framework test cases.
- Pass Rate: 100% Pass Rate (36/36 tests passing cleanly across both frameworks).
- Execution Efficiency: Automated execution reduced total regression cycle time from 18 minutes (1,080 seconds) down to 1 minute 50 seconds (110.58 seconds) — an 89.8% reduction in execution time (9.8x faster).
- Continuous Integration (CI/CD): Fully configured with both Jenkins (`ci_cd/Jenkinsfile`) and GitHub Actions (`.github/workflows/automation-tests.yml`).

2. Test Objectives & Scope

2.1 Primary Objectives

- Transition 100% of manual functional test cases (TC-01 to TC-14) into maintainable, automated test scripts.
- Implement the Page Object Model (POM) pattern to decouple UI element locators from test logic.
- Support both Headed (visible browser) and Headless (CI/CD optimized) execution modes.
- Implement robust explicit synchronization to eliminate test flakiness and handle dynamic React Virtual DOM updates.
- Automate edge-case assertions and reproduce Phase 2 defect scenarios (D-01 and D-02).
- Integrate the test suite into a continuous integration pipeline with automated HTML and JUnit XML reporting.

2.2 In-Scope Functional Modules

- Module 1: Authentication & Authorization (FR-01, FR-02, FR-14 / TC-01, TC-02, TC-14)
Code References: `automation_selenium/pages/login_page.py`,
`automation_selenium/tests/test_authentication.py`
- Module 2: Product Catalog & Sorting (FR-03, FR-04, FR-05 / TC-03, TC-04, TC-05)
Code References: `automation_selenium/pages/inventory_page.py`,
`automation_selenium/pages/product_details_page.py`,
`automation_selenium/tests/test_inventory.py`
- Module 3: Shopping Cart Operations (FR-06, FR-07, FR-08 / TC-06, TC-07, TC-08)
Code References: `automation_selenium/pages/cart_page.py`,
`automation_selenium/tests/test_cart.py`
- Module 4: Checkout Process & Mathematical Calculations (FR-09, FR-10, FR-11, FR-12, FR-13 / TC-09 to TC-13)
Code References: `automation_selenium/pages/checkout_step_one_page.py`,
`automation_selenium/pages/checkout_step_two_page.py`,
`automation_selenium/pages/checkout_complete_page.py`,
`automation_selenium/tests/test_checkout.py`
- Module 5: End-to-End User Purchase Lifecycle (Comprehensive multi-step user journey)
Code Reference: `automation_selenium/tests/test_e2e_workflow.py`

3. Automation Framework Architecture (Page Object Model)

The automated framework is structured following industry-standard Page Object Model (POM) architecture principles, providing high modularity, readability, and single-point locator maintenance.

```
Assignment3/
├── automation_selenium/           # Selenium WebDriver Architec
│   └── automation_robot/         # Robot Framework Test Suite
│       ├── resources/          # Variables, URLs, locators,
│           ├── variables.resource
│           └── test_data
│               └── common.resource
├── config/                       # App URLs, timeouts, test cr
│   └── config.py
├── credentials, directories
├── pages/                        # Page Object Classes (Encaps
│   └── base_page.py            # Master parent wrapper (wait
│       ├── login_page.py      # Login form & error banner l
│       ├── inventory_page.py  # Catalog grid, sorting & car
│       ├── product_details_page.py # Product detail specs & back
│       ├── navigation (FR-05)
│       ├── cart_page.py       # Cart review, item removal &
│       ├── checkout button (FR-07, FR-08)
│       ├── checkout_step_one_page.py # Customer info inputs & vali
│       ├── checkout_step_two_page.py # Order overview, subtotal, 8%
│       ├── tax & total (FR-11)
│       └── checkout_complete_page.py # Confirmation screen & home
├── tests/                        # Pytest Test Suites (Asserti
│   ├── conftest.py             # Browser fixtures, CLI hooks,
│   ├── test_authentication.py  # TC-01, TC-02 (4 combination
│   ├── test_inventory.py       # TC-03, TC-04 (4 sort variat
│   ├── test_cart.py            # TC-06, TC-07, TC-08 (Cart o
│   ├── test_checkout.py        # TC-09 to TC-13 (Checkout &
│   └── test_e2e_workflow.py    # Full End-to-End purchase li
├── automation_robot/
│   ├── resources/
│       ├── variables.resource
│       └── test_data
│           └── common.resource
```

```

s
|   └─ tests/
|       └─ saucedemo_suite.robot           # Automated test cases TC-01
to TC-14
|   └─ ci_cd/                             # CI/CD Pipeline Definitions
|       └─ Jenkinsfile                   # Declarative Jenkins CI pipe
line
|   └─ .github/workflows/automation-tests.yml # GitHub Actions CI workflow
|   └─ reports/                          # Test Artifacts & Visual Rep
orts
|   └─ selenium_test_report.html          # Interactive visual Pytest H
TML report
|   └─ junit_selenium.xml                 # JUnit XML report for CI tes
t parsing
|   └─ robot_logs/                       # Robot Framework report.html
    & log.html
|   └─ run_tests.py                       # Unified CLI test runner
|   └─ pytest.ini                         # Pytest configuration & mark
ers
|   └─ requirements.txt                   # Project dependencies

```

4. Test Execution Summary & Detailed Results

4.1 Quantitative Results Overview

Framework	Total Test Cases / Scenarios	Passed	Failed	Pass Rate	Execution Duration
Selenium WebDriver (Pytest)	22	22	0	100%	110.58s (01:50 min)
Robot Framework	14	14	0	100%	45.22s (00:45 min)
Consolidated Total	36	36	0	100%	155.80s (02:35 min)

4.2 Comprehensive Traceability Matrix & Execution Details

Test Case ID	Requirement	Feature Area	Automated Test Function & Code Location	Execution Result	Execution Time	Automated Verification Highlights
TC-01	FR-01	Authentication	test_authentication.py: test_tc01_successful_login	PASSED	1.42s	Validates redirection to /inventory.html, header display, and catalog load.
TC-02-A	FR-02	Authentication	test_authentication.py: test_tc02_login_error_validation[invalid]	PASSED	1.15s	Asserts "Username and password do not match any user in this service".
TC-02-B	FR-02	Authentication	test_authentication.py: test_tc02	PASSED	1.10s	Asserts "Sorry, this user has

Test Case ID	Requirement	Feature Area	Automated Test Function & Code Location	Execution Result	Execution Time	Automated Verification Highlights
			_login_error_validation[locked]			been locked out.”
TC-02-C	FR-02	Authentication	test_authentication.py: test_tc02_login_error_validation[empty_user]	PASSED	1.08s	Asserts “Username is required”.
TC-02-D	FR-02	Authentication	test_authentication.py: test_tc02_login_error_validation[empty_pass]	PASSED	1.09s	Asserts “Password is required”.
TC-03	FR-03	Inventory	test_inventory.py: test_tc03_inventory_display	PASSED	1.35s	Verifies exactly 6 products, item titles, prices (\$7.99 to \$49.99), and images.
TC-04-A	FR-04	Sorting	test_inventory.py: test_tc04_product_sorting[a-z]	PASSED	1.62s	Verifies alphabetical ordering (A to Z) on catalog.
TC-04-B	FR-04	Sorting	test_inventory.py: test_tc04_product_sorting[z-a]	PASSED	1.58s	Verifies reverse alphabetical ordering (Z to A) on

Test Case ID	Requirement	Feature Area	Automated Test Function & Code Location	Execution Result	Execution Time	Automated Verification Highlights
						catalog.
TC-04-C	FR-04	Sorting	test_inventory.py: test_tc04_product_sorting[low]	PASSED	1.60s	Verifies ascending price ordering (\$7.99 -> \$49.99).
TC-04-D	FR-04	Sorting	test_inventory.py: test_tc04_product_sorting[high]	PASSED	1.61s	Verifies descending price ordering (\$49.99 -> \$7.99).
TC-05	FR-05	Product Details	test_inventory.py: test_tc05_product_details_view	PASSED	1.48s	Validates item title, description, \$29.99 price, image, and back navigation.
TC-06	FR-06	Shopping Cart	test_cart.py: test_tc06_add_to_cart	PASSED	1.75s	Validates live cart badge incrementing from 0 -> 1 -> 2.
TC-07	FR-07	Shopping Cart	test_cart.py: test_tc07_remove_from_cart	PASSED	1.82s	Validates live cart badge decrementing and button label state toggle.
TC-08	FR-08	Cart Review	test_cart.py:	PASSED	1.55s	Verifies /cart.htm

Test Case ID	Requirement	Feature Area	Automated Test Function & Code Location	Execution Result	Execution Time	Automated Verification Highlights
			test_tc08_cart_review			1 item list, prices, and continue shopping navigation.
TC-09	FR-09	Checkout Info	test_checkout.py: test_tc09_checkout_empty_fields_validation	PASSED	1.38s	Submits blank form; asserts "Error: First Name is required".
TC-10-A	FR-10	Input Rules	test_checkout.py: test_tc10_checkout_input_validation[last_name]	PASSED	1.41s	Missing last name; asserts "Error: Last Name is required".
TC-10-B	FR-10	Input Rules	test_checkout.py: test_tc10_checkout_input_validation[postal_code]	PASSED	1.40s	Missing postal code; asserts "Error: Postal Code is required".
TC-11	FR-11	Order Overview	test_checkout.py: test_tc11_order_overview_calculations	PASSED	1.70s	Programmatic formula check: Subtotal (\$45.98) + Tax (\$3.68) == Total

Test Case ID	Requirement	Feature Area	Automated Test Function & Code Location	Execution Result	Execution Time	Automated Verification Highlights
						(\$49.66).
TC-12	FR-12	Order Finish	test_checkout.py: test_tc12_finish_order_confirmation	PASSED	1.65s	Validates “Thank you for your order!” confirmation and pony express image.
TC-13	FR-13	Checkout Cancel	test_checkout.py: test_tc13_cancel_checkout_redirection	PASSED	1.52s	Step 1 cancel returns to Cart; Step 2 cancel returns to Inventory.
TC-14	FR-14	User Logout	test_authentication.py: test_tc14_secure_logout	PASSED	1.88s	Validates sidebar logout, session termination, and unauthorized back-navigation block.
E2E	All	E-Commerce	test_e2e_workflow.py: test_complete_e2e_shopping_workflow	PASSED	3.45s	Full 13-step purchase journey from login to order confirmation to logout.

5. Continuous Integration (CI/CD) Implementation

Continuous testing is integrated using industry-standard CI/CD pipeline definitions:

5.1 Jenkins Pipeline (`ci_cd/Jenkinsfile`)

- Declarative Multi-Stage Pipeline:
 1. Checkout: Pulls latest code from version control.
 2. Setup Environment: Creates virtual environment and installs dependencies from `requirements.txt`.
 3. Run Selenium Pytest Suite: Executes headless Pytest suite and outputs `selenium_test_report.html` and `junit_selenium.xml`.
 4. Run Robot Framework Suite: Executes headless Robot suite and outputs `report.html` and `log.html`.
 5. Archive Test Results: Publishes JUnit XML test results and archives HTML artifacts for stakeholder review.

5.2 GitHub Actions Workflow (`.github/workflows/automation-tests.yml`)

- Automated Trigger: Runs automatically on every push and pull_request to main/master branches.
- Matrix Build: Tested across Python 3.11 with Google Chrome in headless mode.
- Artifact Upload: Automatically uploads HTML test reports and failure screenshots as GitHub workflow run artifacts.

6. Discrepancies & Comparison with Phase 2 (Manual Testing)

Dimension	Manual Testing (Phase 2)	Automated Testing (Phase 3)	Analysis & Impact
Total Test Duration	18 minutes (~1,080s)	1 min 50 sec (110.58s)	89.8% Time Reduction (9.8x Faster)
Human Labor Cost	High (Requires human tester for each run)	Zero (Automated bot execution)	100% labor savings on regression testing
Execution Consistency	Vulnerable to human fatigue & missed steps	100% deterministic & repeatable	Eliminates human execution error entirely
Defect Reproducibility	Manual notes & handwritten steps	Programmatic test scripts & exact assertions	Instant automated defect regression checking
Report Generation	Manual documentation in Word	Instant self-contained interactive HTML & XML	Real-time test visibility for stakeholders

7. Analysis of Phase 2 Defects in Automation

7.1 Defect D-01: Empty Cart Checkout (Severity: High)

- Description: In Phase 2, testers discovered that navigating to `/cart.html` with 0 items displays an active Checkout button, allowing users to proceed through the entire checkout flow.
- Automated Handling: Automated test scripts in `automation_selenium/tests/test_cart.py` and `automation_selenium/tests/test_checkout.py` assert cart state constraints (`assert cart_page.get_cart_item_count() > 0`). In standard regression, this defect scenario is isolated and verified programmatically.
- Development Recommendation: Disable the #checkout button when `.cart_item` count is zero, and display a user-friendly empty cart message.

7.2 Defect D-02: Weak Input Validation (Severity: Medium)

- Description: In Phase 2, checkout customer information fields accepted single-character inputs ("a", "1", "!") without length or postal format rules.
- Automated Handling: Parameterized automated test cases in `automation_selenium/tests/test_checkout.py` (`test_tc10_checkout_input_validation`) systematically assert required field boundaries and document the application's permissive validation logic.
- Development Recommendation: Implement regex format validation on postal codes (e.g., Canadian K1A 0B1 / US 90210) and minimum 2-character constraints on first and last names.

8. Conclusion & References

8.1 Project Conclusion

The CST8513 Phase 3 Automation Testing project for SauceDemo has been completed with 100% success. By implementing the Page Object Model (POM) architecture using Selenium WebDriver and Robot Framework, all 14 functional requirements (FR-01 to FR-14) have been fully automated, achieving: - 100% test pass rate across all 36 test scenarios. - 89.8% reduction in regression execution time. - Seamless CI/CD readiness with Jenkins and GitHub Actions.

8.2 Technical References & Artifacts

1. Course Syllabus & Instructions: CST8513 Assignment III Specifications, Prof. Sharmista Datta, Algonquin College.
2. Phase 1 Project Proposal: CST8513 Project 1 Requirements Analysis, Devangbhai Pandit & Meet Ahalpara.
3. Phase 2 Test Cases & Defect Report: CST8513 Project 2 Test Cases & Defects Report, Devangbhai Pandit & Meet Ahalpara.

References

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